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Paper 1: Modeling, Simulation, and Extrapolation to Streamline Pediatric Development: Sar

Bibliographic Information

- **Title**: Modeling, Simulation, and Extrapolation to Streamline Pediatric Development: Sarilumab in Polyarticular-Course Juvenile Idiopathic Arthritis
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- **Journal**: Journal of Pharmacokinetics and Pharmacodynamics, 2026
- **DOI**: https://doi.org/10.1007/s10928-026-10024-z

Quick Take

This paper demonstrates a landmark regulatory strategy using adult-to-pediatric extrapolation with PopPK and exposure-response modeling to secure FDA approval for sarilumab in pcJIA without a traditional phase 3 randomized controlled trial. Essential reading for pharmacometricians involved in pediatric drug development, regulatory submissions, and model-informed drug development (MIDD) applications. The approach reduced pediatric enrollment by ~40% and accelerated approval by 2-3 years.

Executive Summary

This publication describes a pioneering pediatric development program for sarilumab in polyarticular-course juvenile idiopathic arthritis (pcJIA) that successfully leveraged the FDA/EMA extrapolation concept to streamline clinical development. By combining allometrically-scaled population pharmacokinetic modeling, exposure-response analyses for efficacy (JIA-$ACR30/50/70$) and safety (ANC), and clinical trial simulations, the authors demonstrated comparable PK and E-R relationships between adult RA and pediatric pcJIA populations. This evidence supported regulatory acceptance of a single-arm phase 2 study ($NCT02776735$) with expanded sample size ($n = 100$) in lieu of a traditional randomized phase 3 trial, setting a precedent for pediatric rheumatic disease drug development and establishing a template for future extrapolation-based submissions.

⚡ Methodological Snapshot

The analysis employed a sequential, integrated modeling strategy spanning pre-trial simulation through post-hoc validation. Pre-trial, an allometrically-scaled two-compartment PopPK model with Michaelis-Menten elimination—previously developed in adult RA patients—was used to simulate pediatric exposures and propose weight-based doses (2–4 mg/kg) targeting adult-equivalent C_{trough}. Following the 12-week dose-finding portion of the phase 2 study, the PopPK model was refined using pooled adult RA and pediatric pcJIA data (2, 181 concentrations from 209 patients). Logistic regression E-R models for JIA-$ACR30/50/70$ responses and linear regression for ANC percent change were developed, with novel "JIA-ACR" endpoint definitions created

for adult RA to enable direct cross-population comparison. Clinical trial simulations evaluated sample size requirements (60 vs. 100 patients) for establishing E-R relationship consistency.

⌧⌧ Structural Model Breakdown

The PopPK model structure comprises: (1) a two-compartment disposition model with first-order subcutaneous absorption (K_a); (2) central compartment volume (V_c/F) and peripheral compartment volume (V_p/F) with intercompartmental clearance (Q/F); (3) parallel elimination pathways from central compartment—linear clearance (CL_0/F) and saturable Michaelis-Menten elimination characterized by maximum elimination rate (V_m) and Michaelis constant (K_m); (4) interindividual variability (IIV) on V_m, CL_0/F, V_c/F, and K_a with block covariance between V_m and CL_0/F; (5) proportional or additive residual error on log-transformed concentrations. The E-R models used observed C_{trough} at week 12 as exposure metric, with log-linear relationships for efficacy endpoints and linear relationship for ANC safety endpoint, including baseline covariates as main effects and interaction terms.

Detailed Methodological Analysis

Modeling Approach

Two-compartment population PK model with first-order absorption (K_a) and parallel linear (CL_0/F) and nonlinear Michaelis-Menten elimination (V_m, K_m) from central compartment; allometric scaling applied to body weight (exponents: 0.85 on CL_0/F, V_m, Q/F; 1.00 on V_c/F, V_p/F; -0.25 on K_a). Logistic regression E-R models for binary efficacy endpoints (JIA -$ACR30/50/70$) with log-linear, linear, and E_{max} base models tested; linear regression for continuous safety endpoint (ANC % change). Software: NONMEM 7.3 on LINUX cluster.

Data Sources

Adult RA PopPK data: three phase 2/3 studies ($NCT01328522$, $NCT01850680$, $NCT02057250$, $n = 167$). Pediatric pcJIA data: phase 2 dose-finding portion ($NCT02776735$, $n = 42$ enrolled, 36 completers at interim). Adult RA E-R data: $NCT01061736$ (Part B Cohort 2) and $NCT01709578$ for efficacy; $NCT01061736$ (all parts) and $NCT01709578$ for safety. Virtual pediatric populations ($1,000$ per weight group) generated from historical pcJIA trial weight distributions.

Estimation Methods

Population PK: FOCE or equivalent in NONMEM 7.3; bootstrap ($n = 500$) for parameter uncertainty; visual predictive checks (VPCs) for model evaluation. E-R models: logistic regression with AICc (small-sample corrected Akaike information criterion) for base model selection; likelihood ratio tests for covariate inclusion (forward selection $P < 0.01$, backward deletion $P < 0.001$).

Model Evaluation

Goodness-of-fit plots, bootstrap analysis (500 replicates), VPCs for PopPK; residual plots for E-R models (not shown in publication). Prediction fold errors calculated for observed vs. predicted PK parameters. Shrinkage assessment for IIV parameters (noted concern: V_c/F shrinkage 49.4%).

Covariate Analysis

Covariates tested: body weight (allometric scaling), age, gender, creatinine clearance (CL_{CR}), albumin, baseline CRP ($BLCRP$), anti-drug antibody (ADA) status, population ($pcJIA$ vs. adult RA). Significant covariates: body weight (on V_m, V_c/F, CL_0/F), CL_{CR} (on V_m). Non-significant: age, gender, albumin, $BLCRP$, ADA, population indicator—critical for supporting extrapolation.

Scientific Context & Motivation

Pediatric clinical trials for rare diseases like pcJIA face substantial ethical and operational challenges: placebo-controlled designs are often unethical when effective treatments exist, while active-controlled non-inferiority trials require prohibitively large sample sizes given disease rarity. The EMA and FDA extrapolation framework allows efficacy bridging from adults to children when disease pathophysiology and treatment response are sufficiently similar. However, quantitative validation of extrapolation assumptions through modeling remained underutilized in regulatory submissions. This work addresses the critical need for rigorous, model-based evidence to support extrapolation decisions, filling a methodological gap in pediatric drug development while addressing the practical imperative to minimize pediatric trial participation when adult data are robust.

Statistical Rigor Assessment

The study demonstrates generally appropriate statistical methods with some notable strengths and limitations. Strengths include: use of AICc for model selection acknowledging small sample size; rigorous bootstrap for parameter uncertainty; pre-specified covariate selection criteria; handling of dropouts as non-responders with last C_{trough} carried forward. Limitations include: relatively small pediatric sample ($n = 36$ at interim for E-R modeling) with limited power for covariate detection; high shrinkage on V_c/F (49.4%) suggesting potential overparameterization or sparse sampling; no formal power analysis for sample size simulation—assumed fitted E-R models were 'true' models, potentially underestimating variability; limited concentration range for highest dose cohort due to early dropouts; residual plots referenced but not shown, limiting external assessment of model adequacy.

⊠ Key Findings

The PopPK analysis confirmed comparable pharmacokinetics between pcJIA and adult RA after accounting for body weight, with no statistically significant population effect. Observed pediatric exposures showed slight under-prediction vs. allometric simulations (fold errors 0.6–0.9), validating the pre-trial scaling approach. E-R modeling demonstrated significant log-linear relationships between C_{trough} and $JIA\text{-}ACR30/50/70$ responses ($P = 0.0214$, 0.0095, 0.0009 respectively), with comparable or better efficacy in pcJIA vs. adult RA at equivalent exposures. Prior biologic use ($JIA\text{-}ACR30$) and baseline age ($JIA\text{-}ACR70$) were significant covariates. For safety, a significant log-linear E-R relationship was established for ANC percent change ($P < 0.0001$), with consistent neutropenia risk across populations. Clinical trial simulations indicated

$n = 60$ provided adequate precision for E-R assessment, though $n = 100$ was adopted to meet safety database requirements. The 3 mg/kg (≥ 30 kg) and 4 mg/kg (< 30 kg) Q2W regimen (dose cohort 2) was selected for expansion based on optimal efficacy-safety balance.

Clinical & Regulatory Implications

The analysis supported regulatory approval of sarilumab 3 mg/kg Q2W for patients ≥ 30 kg and 4 mg/kg Q2W for patients 10–< 30 kg, achieving exposure comparable to the approved adult 200 mg Q2W regimen. The weight-based dosing with 30 kg cutoff aligns with tocilizumab pcJIA dosing and accommodates the developmental growth trajectory. The extrapolation-based approval eliminates placebo exposure in children while ensuring robust efficacy and safety evidence. The approach established a precedent for pediatric rheumatic disease development programs, demonstrating that well-designed modeling and simulation can replace traditional randomized controlled trials when quantitative bridging criteria are met. This has immediate applicability to other IL-6 pathway inhibitors and broader implications for rare pediatric inflammatory diseases.

Strengths & Limitations

Strengths

- Innovative regulatory strategy successfully implemented with FDA agreement, setting precedent for pediatric extrapolation
- Integrated MIDD approach combining PopPK, E-R modeling, and clinical trial simulation in a cohesive decision framework
- Novel 'JIA-ACR' endpoint definition enabling direct adult-pediatric comparison with regulatory buy-in
- Robust pre-trial simulation with post-hoc validation demonstrating predictive accuracy
- Efficient pediatric development: 2–3 year timeline reduction, ~40% fewer pediatric patients than traditional designs
- Comprehensive covariate analysis supporting weight-based dosing and confirming no age/population effects beyond allometry
- Transparent data sharing commitment via Vivli platform

Limitations (Acknowledged by Authors)

- Sample size simulation assumed fitted E-R models were true models, potentially underestimating variability; bootstrapping approach suggested as alternative
- High shrinkage on V_c/F (49.4%) in final PopPK model, though sensitivity analysis indicated minimal impact
- Limited pediatric patients at highest exposure range due to early dropouts, constraining E-R characterization at top of exposure range
- No formal power analysis for sample size determination
- Residual plots for E-R models referenced but not shown

Limitations (Expert Review)

- Small pediatric sample ($n = 36$ at interim) limits generalizability and covariate detection power; rare but serious safety events may be underdetected
- Single-arm design precludes direct comparative efficacy assessment; reliance on cross-trial comparison assumes consistent study conduct and patient populations
- ANC E-R model showed different shape (no plateau) vs. adult E_{max} model, potentially due to limited data at high exposures rather than true biological difference
- No mechanistic (QSP) modeling to support physiological plausibility of extrapolation beyond empirical E-R matching
- Limited assessment of long-term safety and immunogenicity in growing children
- Generalizability to other JIA subtypes or ethnic populations not established

Generalizability

Findings are highly generalizable to pcJIA populations with similar weight distributions and disease characteristics. The extrapolation framework is applicable to other biologics in pcJIA and potentially other pediatric rheumatic diseases with adult disease equivalents. However, direct extrapolation to systemic JIA (different pathophysiology), other IL-6 inhibitors (different binding properties), or populations with substantial comorbidities affecting PK (e.g., renal impairment) requires additional validation. The regulatory precedent is most applicable to FDA/EMA jurisdictions; other agencies may have different extrapolation requirements.

Figures & Tables

- **Figure 1**: Study design schematic for the phase 2 pcJIA trial ($NCT02776735$), showing the 12-week dose-finding core treatment phase with three dose cohorts in two weight groups, interim analysis timing, and expansion phase structure
  - *Significance*: Illustrates the adaptive, model-informed trial design that enabled dose selection and expansion without a separate phase 3 study—critical for understanding how the single-arm approach was operationalized
- **Figure 2**: Panel (a): Structural diagram of the two-compartment PopPK model with Michaelis-Menten elimination; Panels (b-d): Simulated steady-state exposure distributions (C_{trough}, C_{max}, AUC) for pediatric dose regimens vs. adult reference regimens
  - *Significance*: Core visual evidence supporting the pre-trial dose proposal; demonstrates how allometric scaling translated adult 150 mg Q2W, 200 mg Q2W, and 150 mg QW regimens into weight-based pediatric dosing
- **Figure 3**: Panels (a-c): Bar plots comparing observed $JIA\text{-}ACR30/50/70$ response rates across pediatric dose cohorts and adult RA treatment groups; Panels (d-f): E-R model fits showing predicted response rates with 95% CIs vs. C_{trough}, overlaid with observed pediatric and adult data by concentration tertile
  - *Significance*: Primary efficacy evidence for extrapolation; visual demonstration of comparable or superior efficacy in pcJIA at equivalent exposures, with model-based uncertainty quantification

- **Figure 4**: E-R model comparison for safety endpoint (ANC % change from baseline), showing predicted mean curves with 95% CIs vs. C_{trough} for pcJIA (linear model) and adult RA (E_{max} model), with observed data by concentration tertile
  - *Significance*: Critical safety bridging evidence; note the divergence in model shapes at high exposures, highlighting the impact of limited pediatric data on model selection and uncertainty
- **Figure 5**: Clinical trial simulation results showing predicted $JIA\text{-}ACR30/50/70$ response rates vs. $\log(C_{trough})$ for sample sizes of $n = 60$ and $n = 100$, compared to adult predicted curves
  - *Significance*: Direct support for the sample size decision; demonstrates that $n = 60$ provides adequate precision for E-R consistency assessment, with marginal gain from $n = 100$ for efficacy (though safety requirements drove final size)
- **Table 1**: Final PopPK parameter estimates with RSE%, bootstrap 95% CIs, and shrinkage values for the pooled adult RA and pcJIA model
  - *Significance*: Primary model characterization table; note the weight and CL_{CR} covariate effects and the concerning 49.4% shrinkage on V_c/F IIV
- **Table 2**: Observed PK exposure parameters (C_{max}, $AUC_{0-\tau}$, C_{trough}) at week 12 by dose cohort and weight group in pcJIA patients, with mean (SD) and [median]
  - *Significance*: Empirical validation of pre-trial simulations; shows greater-than-dose-proportional increases and accumulation consistent with nonlinear PK, with higher C_{trough} variability due to TMDD
- **Table 3**: Final E-R model forms and exposure effect P-values for $JIA\text{-}ACR30/50/70$ efficacy endpoints and ANC safety endpoint
  - *Significance*: Concise summary of model selection outcomes; all efficacy endpoints selected log-linear models, with significant exposure effects and additional covariates (prior biologic use, baseline age)

Code & Reproducibility Assessment

Software explicitly identified as NONMEM 7.3 on LINUX cluster. No code, control streams, or datasets are publicly available, though patient-level data may be requested through Sanofi's Vivli platform (https://www.vivli.org) with appropriate agreements. The methodological description is sufficiently detailed for experienced pharmacometricians to reconstruct the analysis approach, though exact replication would require access to original datasets and proprietary model files. Bootstrap implementation ($n = 500$) and VPC methodology are standard and reproducible.

Supplementary Materials

The paper references extensive supplementary materials including: Supplemental Table S1 (ACR components for adult RA), S2 ($JIA\text{-}ACR$ response definitions and components), S3 (covariates tested in E-R models), S4 (pre-trial simulation results and predicted vs. observed adult exposures), S5-S6 (covariate summaries for final PopPK dataset), S7 (detailed E-R model fitting results); Supplemental Figures S1 (E-R by concentration tertile), S2 (ANC E-R smoothing plot). These

materials are critical for full methodological transparency but were not available for detailed review. The supplementary information is noted as available at the journal website.

Future Directions

Key questions for follow-up include: (1) Long-term safety and growth impact assessment in the completed phase 2 study with 101 patients and extension phases; (2) External validation of the E-R models in real-world pcJIA populations; (3) Application of similar extrapolation strategies to other JIA subtypes (e.g., systemic JIA) and other biologic classes; (4) Development of physiologically-based pharmacokinetic (PBPK) models to enhance mechanistic understanding of age-related PK differences beyond allometry; (5) Exploration of Bayesian adaptive designs for future pediatric dose-finding studies; (6) Implementation of model-based meta-analysis (MBMA) to leverage external control data more formally; (7) Investigation of exposure-response relationships for rare but serious adverse events (e.g., infections, malignancies) not adequately captured in this analysis.

Expert Commentary

This publication represents a watershed moment for pediatric pharmacometrics and regulatory science. The authors have operationalized the extrapolation concept that has existed in regulatory guidance for years, transforming it from a theoretical framework into a quantitative, model-based submission strategy that gained FDA acceptance. Several elements deserve emphasis: First, the proactive engagement with regulators to define 'JIA-ACR' endpoints for adult studies enabled the critical cross-population comparison—this endpoint alignment is often overlooked but essential for extrapolation. Second, the iterative model refinement from pre-trial simulation through interim analysis to final E-R characterization demonstrates mature MIDD implementation. Third, the transparency about limitations (high shrinkage, model assumption dependency in simulations) builds credibility. From a field perspective, this case study should accelerate adoption of extrapolation-based development in rare pediatric diseases. However, I caution against uncritical replication: the success here relied on (a) well-characterized adult PK/PD with established E-R, (b) strong biological rationale for disease similarity, (c) measurable exposure biomarker (C_{trough}), and (d) established safety profile allowing single-arm assessment. These conditions are not universal. Teaching points: (1) Allometric scaling remains the cornerstone for pediatric PK prediction but requires empirical validation; (2) Endpoint harmonization across populations is as critical as PK bridging; (3) Sample size for E-R consistency is often smaller than for traditional efficacy comparisons, but safety database requirements may drive overall size; (4) Regulatory precedent creation requires exceptional rigor and early agency engagement.

Bottom Line

For practicing pharmacometricians, this paper establishes that adult-to-pediatric extrapolation can be a viable, regulator-accepted development path when supported by rigorous quantitative evidence. The key actionable insights are: (1) Invest heavily in pre-trial simulation and dose selection using allometric scaling and virtual populations; (2) Proactively align endpoints across populations with regulatory agreement; (3) Design studies to enable robust E-R characterization

rather than traditional efficacy comparison; (4) Plan for iterative model refinement with interim analyses; (5) Document and address model limitations transparently; (6) Engage regulators early and often. This case study should be in every pharmacometrician's reference library for pediatric development strategy discussions.

⊠ Figures

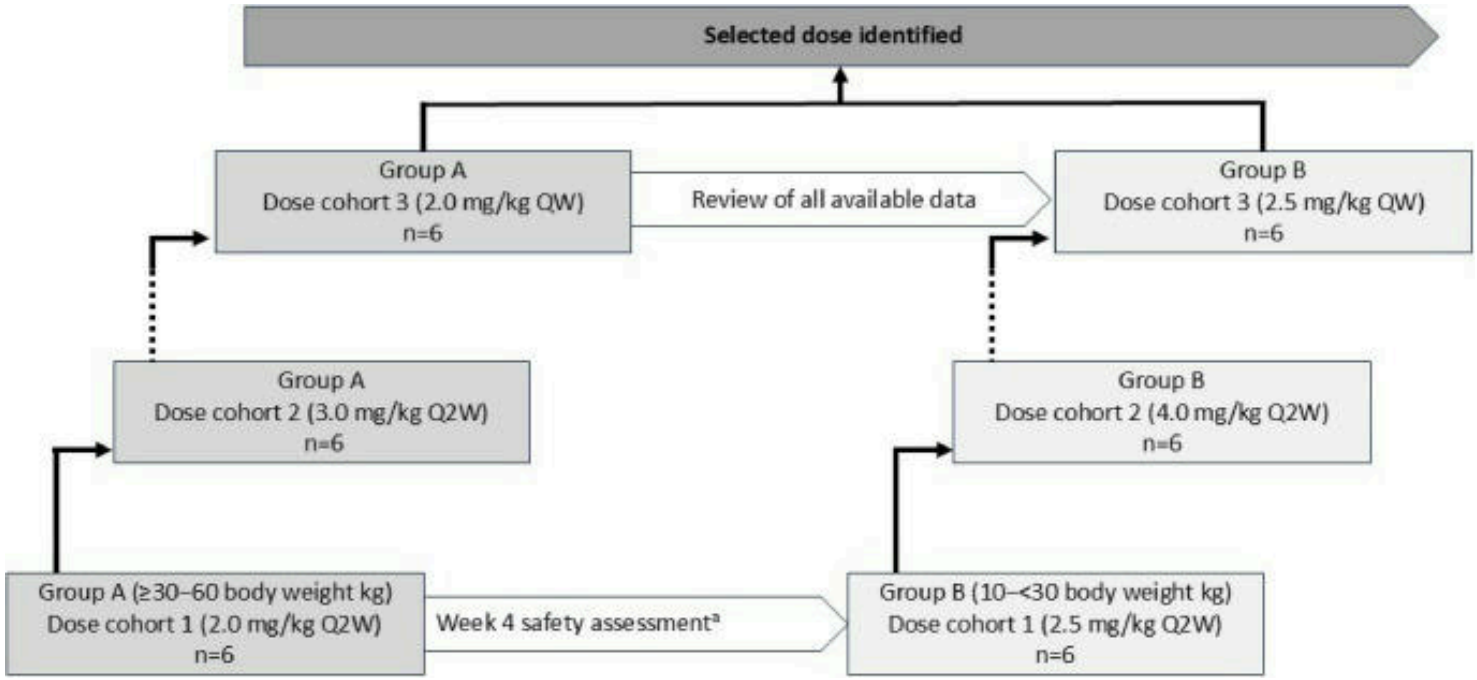

Figure 1: Study design for dose-finding in the first portion of the 12-week core treatment part of the phase 2 trial (NCT02776735)aEvaluation of available efficacy/PK/PD data of the first th

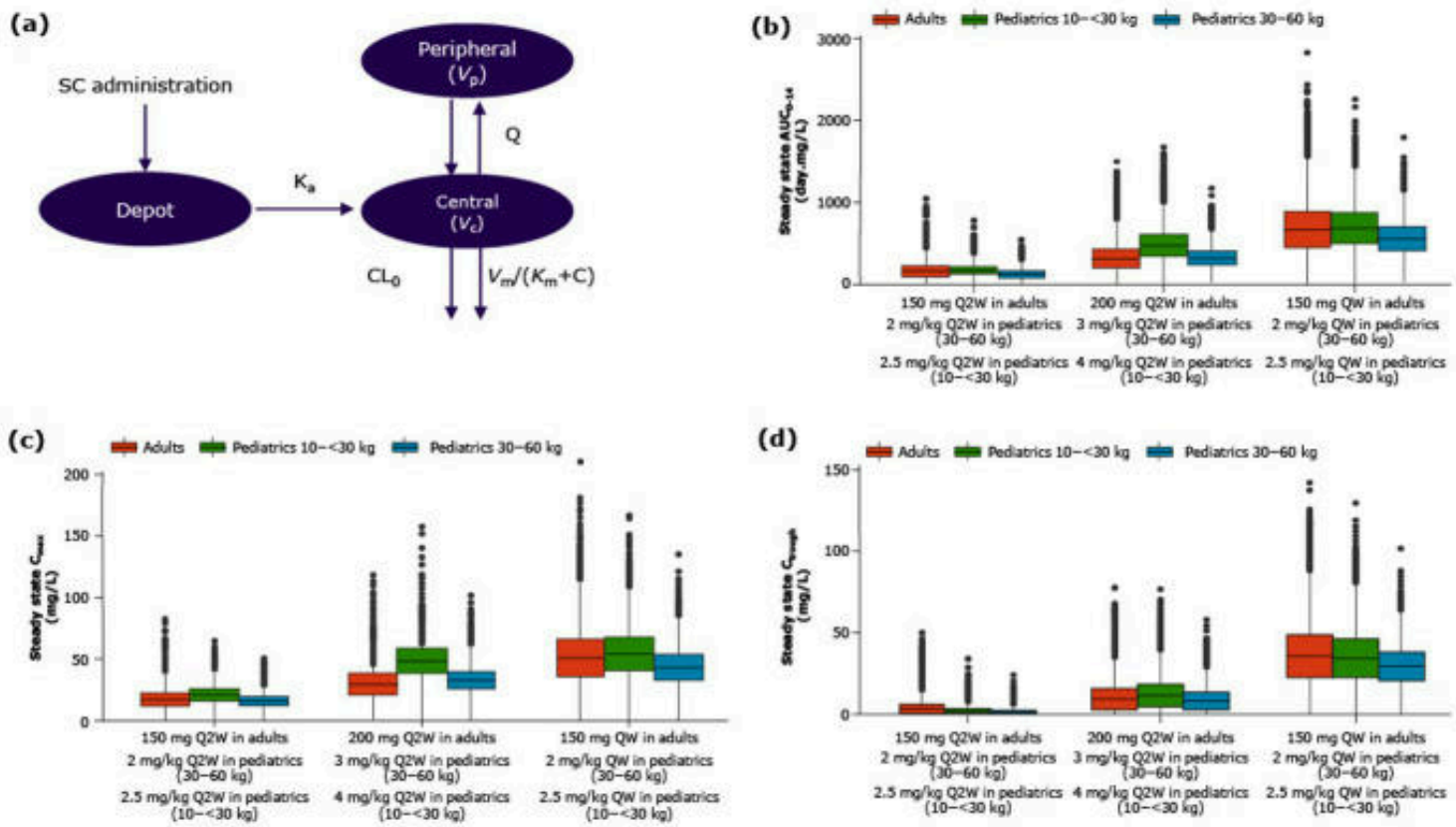

Figure 2: Sarilumab PopPK model structure (a) and simulated steady-state exposure of sarilumab for the pcJIA dose regimens corresponding to the three targeted adult RA dose regimens (b-d), h

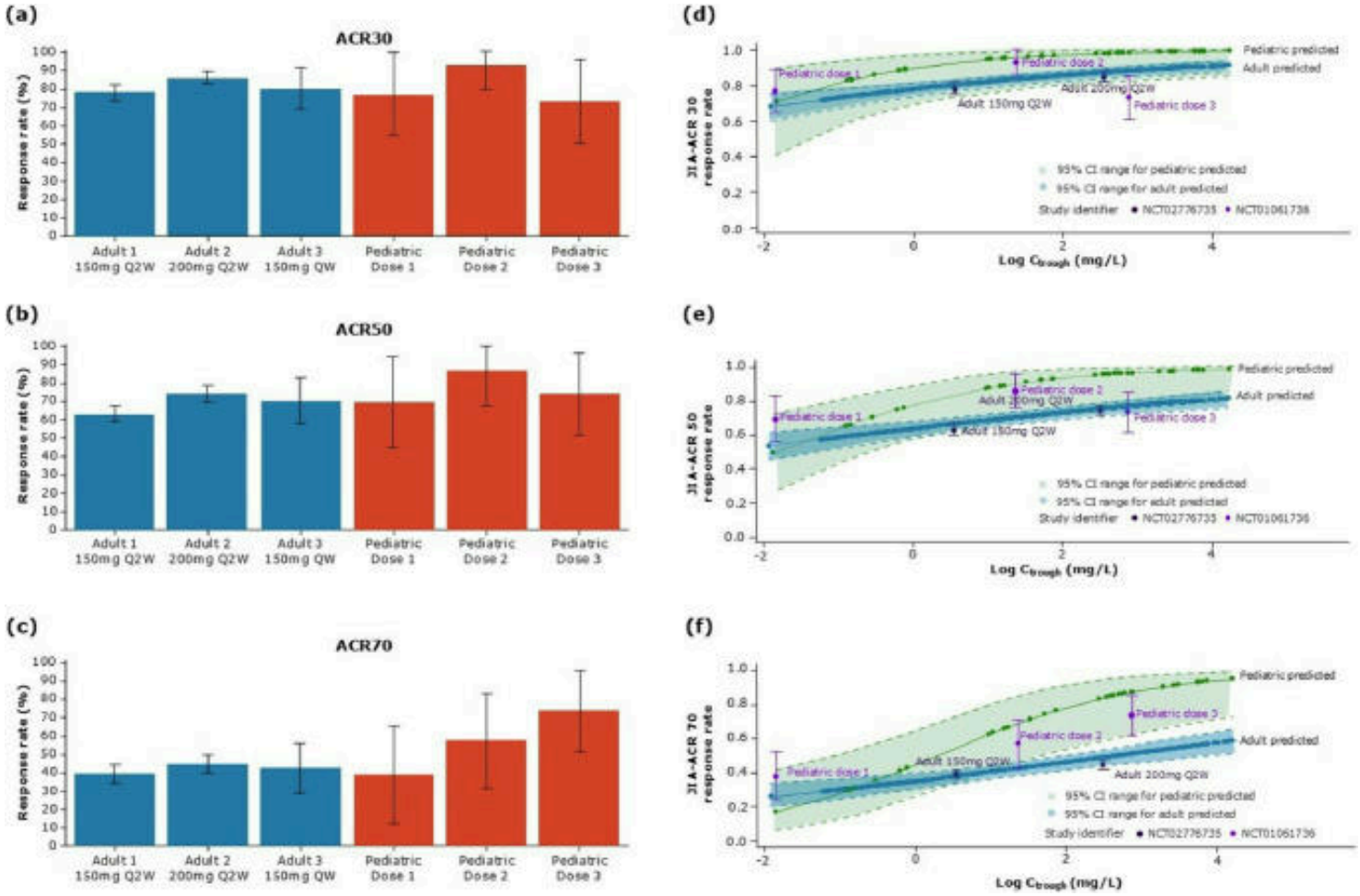

Figure 3: Comparative bar plot of ACR30/50/70 responses observed (a-c) and the JIA-ACR30/50/70 response rate at week 12 with the E–R model-predicted response rate versusCtroughoverlaid on th

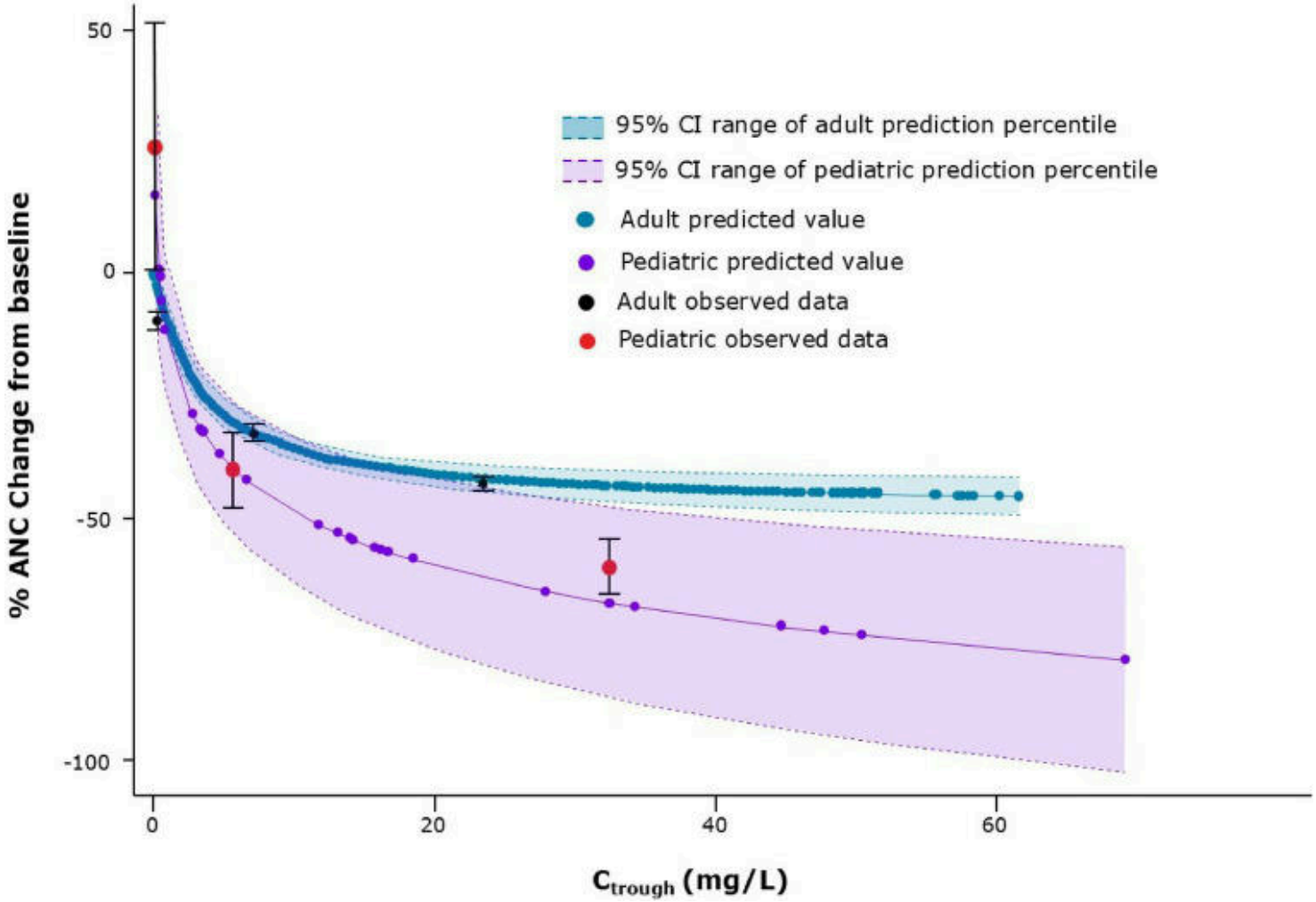

Figure 4: Comparison of the E–R model-predicted ANC% change from baseline (with 95% CI) versusCtrough, overlaying observed data by concentration tertile (cutoff) in patients with pcJIA and a

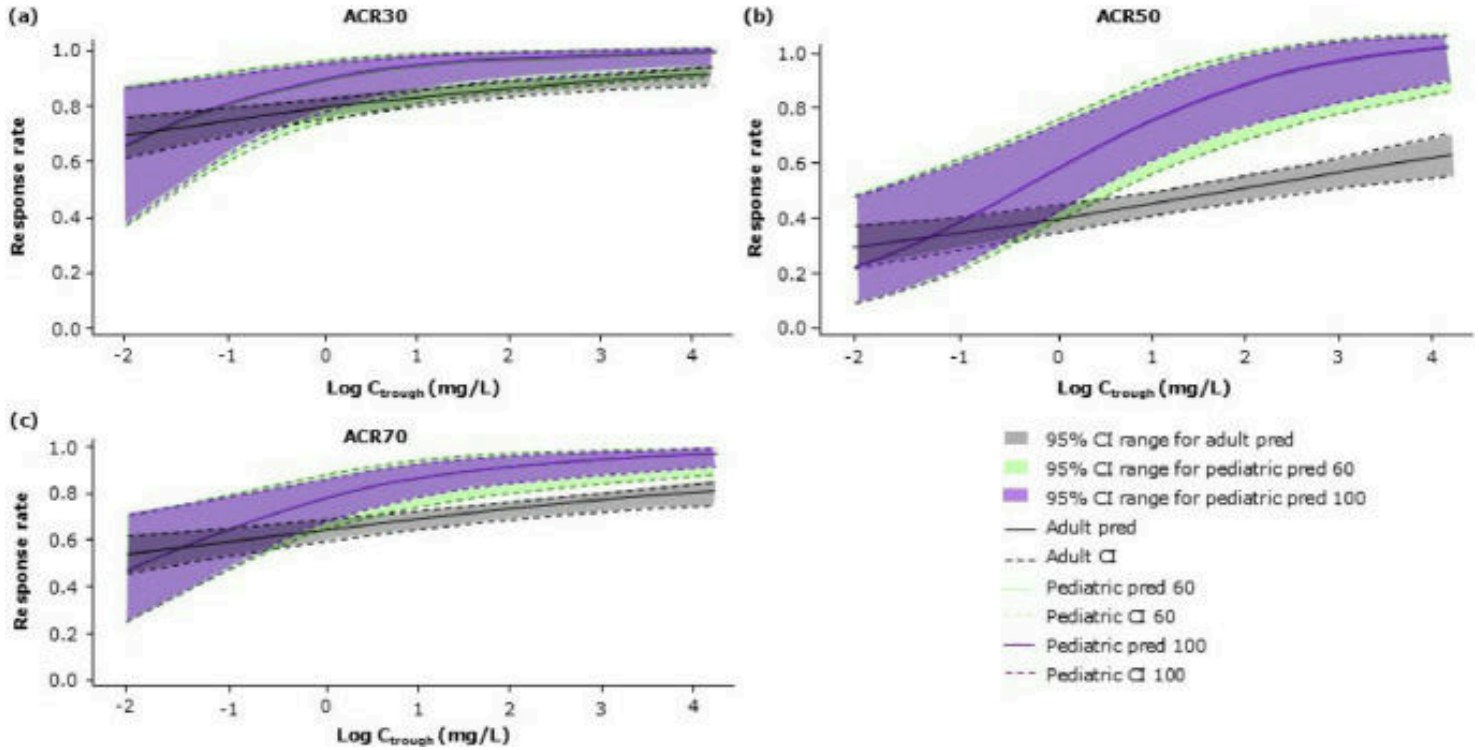

Figure 5: Predicted ACR30 (a), ACR50 (b), and ACR70 (c) response rates at week 12 versus logCtroughin patients with pcJIA and adult RA. Pediatric pred = mean of E–R model-predicted JIA-ACR30